Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	**Name**	Danz, Norbert	**Email**	Norbert.Danz@iof.fraunhofer.de
Institute Name	Fraunhofer IOF			
Address	Albert-Einstein-Str. 7, 07745 Jena, Germany			

2. Technology Overview

Title of Proposed Technology	*Optical device technology based on double-sided maskless irregular microlens arrays for compact headlight, ground-projection and other illumination systems*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), ~~Secondary Battery ( )~~, Advanced Mobility (√), ~~Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )~~			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		*Electrical・Electronics*	
	Medium Category	Small Category	Medium Category	Small Category
	Optical Application Devices	*Optical Devices*	*Home Appliances and Electronic Application Devices*	*Lighting Devices*
Concept of Proposed Technology	*Maskless irregular microlens arrays allow to build very compact illumination and pattern projection optical systems with high design and light distribution freedom as well as very high efficiency due to the absence of absorptive structures. This offers to mass-produce the pattern shapers by large volume replication (injection molding), thus qualifying them for large scale applications e.g. in the automotive business field.* *Potential applications of such systems are:* - *Automotive: static [1] and dynamic headlight systems [2]; static and dynamic pattern and ground projection systems [3]; exterior and interior lighting & display systems [4].* - *Bio-medical: miniature spectroscopic light sources, tailored illumination for analytical instruments (e.g. microscopes, …).*			

(max 10 pages)

	- *General lighting: artificial sun displays; stage lighting; advanced illumination systems.*
Rationale for Joint R&D Project	*Fraunhofer IOF has established design concepts for applications of maskless irregular microlens arrays and proved their feasibility based on in-house "polymer-on-glass" replication technology. Currently, further "closer-to-product" research is required in collaboration with industrial partners (Tier 2…1 or OEMs) for the development of new products/applications and projection functionalities according to specific customer needs. This also includes establishing a workflow from the optical design via manufacturing (by injection molding) to the system integration.* *The project will start with the existing design expertise, applying it to the desired applications / functions, and exploiting IOF's in-house mastering technologies for deriving the replication tools required.*
Key Words	1. *LED beam shaping* 2. *Maskless microlens arrays* 3. *Injection molding* 4. *Irregular fly's eye condensers* 5. *Miniaturized optical systems*

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (4)	**End of the Project**	Level (7)
R&D Objectives	*Design and technological objectives:* - *Advanced optical design of maskless irregular microlens arrays with reduced straylight* - *High-precision mastering of double sided, irregular microlens arrays via direct grayscale photolithography or other suitable technologies* - *Proof-of-function based on polymer-on-glass replicated elements with subsequent optical analysis* - *Advanced high-precision replication via injection molding, embossing or micro-/nano-imprint technologies ensuring adequate front side to back side alignment of the optical structures* *Product development objectives:* - *Develop a functional demonstrator for the automotive (or other) illumination/projection application (e.g., projective turning indicator or reverse driving indicator, enhancing road safety).* - *Transfer design into functional prototype device*			
R&D Target Specifications	- *Cost-effective mass-production of the pattern shapers with <10 µm lateral accuracy of front-to-back side lateral positioning of the molded arrays* - *>85% transmission of the pattern shaping element* - *<3% stray light in the projected pattern*			

(max 10 pages)

R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* *Key tasks with approximated durations:* *Phase I – Pre-study (6 months):* - *Definition of the product specification and requirements – 2M* - *System(s) pre-study and preliminary optical simulations – 4M* *Milestone I:* - *Feasibility of design approach, predicted specs, and estimated fabrication tolerances agreed* *Phase II – Optical design and mastering (18 months)* - *Micro-optics and overall optical system design – 6M* - *Micro-optics mastering – 6M* - *Micro-optics replication – 6M* - *Preparing the tool for molding development* - *Prepare reference structures for establishing the molding process* - *Establish and initially optimize molding by means of reference structures* - *Develop electro-mechanical module design* *Milestone II:* - *Master-samples of maskless irregular microlens array(s) available* - *Lab test of polymer-on glass replicated sample in spec* - *Replication technology established* *Phase III – System integration and validation (12 months):* - *Components fabrication – 3M* - *System integration (e.g., car headlight) – 3M* - *System characterization, field tests and validation – 6M* *Milestone III:* - *Successful MLA replication with sufficient precision* - *Assembled demonstrator and performance review*

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	- *Development of new optical system designs and MLA design approaches for straylight suppression with maskless MLAs adapted to the targeted projections* - *Mastering of MLAs using grayscale lithography; preparing polymer-on-glass replicas of the master structures for proving their functionality and delivering for tooling preparation* - *System tolerancing and characterization (laboratory scale), evaluating inline control options* - *Generation of scientific contributions, such as patents or white papers*
Expected R&R of Korean SME Partner	- *Specify targeted application, define boundary conditions (module sizes etc.)* - *MLA replication: Development and optimization of cost-effective mass production technology (e.g. using embossing or injection molding)* - *Quality control and assurance (industrial scale)* - *Opto-electronic-mechanical system integration* - *System application and validation (field tests)*

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	***Technical Outcomes*** • *Build a micro-optics component platform for maskless irregular MLA-based optical systems (e.g., low beams, adaptive driving beams, road projection systems) by establishing the full workflow of the design & development & manufacturing chain.* • *Improve yield and consistency by integrating shape–optical performance correlation models and inline inspection standards; reduce stray light/glare and enable miniaturized, lightweight modules.* ***Economic Impact*** • *As intelligent headlamps and communication lighting expand, domestic production of high-precision micro-optics will cut lead times, lower supply-chain risk, and improve cost competitiveness.* • *Secure domestic vehicle/module references and use them as a launchpad to grow share in related components/modules through export-oriented expansion (UNECE-compliant).* • *Patent protected optical design approach yielding scalable brightness of the projection by scaling the array size* ***Globalization*** • *Combination of overseas partners' development, mastering & metrology infrastructures with domestic mass production and modularization to shorten the development.* • *Apply UNECE design and test scenarios from the start to strengthen European market entry and improve responsiveness to global OEMs/Tier 1s.* • *Build defensible barriers by packaging core processes and design elements into patents and proprietary know-how.*

6. References

(1) L. M. Wilhelm, D. Stefanidi, P. Schreiber, et.al., "Design of maskless micro-optical automotive low-beam," Proc. SPIE 13019, Optical Design and Engineering IX, 130190O (17 June 2024); https://doi.org/10.1117/12.3017435

Picture 1 – Maskless MLA low beam performance (left) and its proof-of-concept demo (right)

(2) D. Stefanidi, L. M. Wilhelm, P. Schreiber, et.al., "Adaptive LED illumination with maskless irregular lenslet arrays and controlled crosstalk," J. Optical Microsystems 5(2) 024502 (22 April 2025) https://doi.org/10.1117/1.JOM.5.2.024502

Picture 2 – Maskless MLA adaptive driving beam performance (left) and its proof-of-concept demo (right)

(max 10 pages)

(3) R. Kundu, D. Michaelis, D. Stefanidi, et.al., "Performance of UV-molded versus injection molded maskless microlens arrays for multi-aperture projection," Proc. SPIE 13021, Optical Fabrication and Testing VIII, 130210J (17 June 2024); https://doi.org/10.1117/12.3016164

Picture 3 – Maskless MLA turn indicator performance (left) and its proof-of-concept demo (right)

(4) R. Kundu, P. Schreiber, P. Dannberg, et. al., “Design and realization of micro-optic lightfield displays,” DGaO Proceedings 2022; https://www.dgao-proceedings.de/download/123/123_h6.pdf

Picture 4 – MLA-based lightfield displays with direct-view 3D effect

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization